

Answer C

- 40) Every decay that result in α emission, proton number decrease by 2, neutron decrease by 2. Nucleon number drop by 4.
Every decay that result in β emission, proton number increase by 1, neutron decrease by 1. Nucleon number remains the same.
For nuclide Q, $\alpha \beta \beta$ emission will result in proton number ending up the same.
For nuclide R, $\beta \beta \beta$ emission will result in the nucleon number unchanged.

Answer D